MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-BCA (HONOURS)

REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER Core Course (CC) - MG2CCRBCA100 –

Mathematics Foundation To Computer Science

Duration: 2 Hours Maximum Marks: 70

SCHEME OF VALUATION

Part A

Answer any 5 questions. Each question carries 2 marks

1. The graph must be connected and have exactly 2 vertices of odd degree. (1+1 mark)
2. False. A tree is a connected graph with no cycles. (1 mark)
   Counterexample: A cycle of 3 vertices (triangle) is connected but has a cycle, hence not a tree./ Any suitable example (1 mark)
3. Steps(1 mark), Approximate value = 24/7 = 3.4286 (1 mark)
4. Steps(1 mark), $x_{n+1}=x_n-\frac{(x_n^3-x_n-3)}{(3x_n^2-1)}$ (1 mark)
5. The objective function is the linear function of the decision variables that is to be maximized or minimized./ Example (2 marks)
6. $4x_1 + 3x_2 - s_1 + A_1 = 8$ (2 marks)
7. Balanced: Total supply = Total demand.(1 mark)
   Unbalanced: Total supply ≠ Total demand(1 mark)
8. MODI = Modified Distribution method.(1 mark)
   Purpose: To test the optimality of a basic feasible solution in a transportation problem/ If the solution is not optimal, modify the solution.(1 mark)

Part B

Answer any 5 questions. Each question carries 6 marks

9. Connected Graph: A graph is said to be connected if there exists at least one path between every pair of vertices.(1.5 marks)
   Disconnected Graph: At least one pair of vertices has no path.(1.5 marks)
   Disconnected graph with 5 vertices and 3 components (3 marks)
10. Start with 1 vertex(1 mark), Add second vertex(1 mark),
    Then add other vertices one by one without forming a cycle, repeat the process until all the vertices are included (4 marks).
11. Difference table (2 marks)
    Formula (1 mark)
    Newton's Backward Difference at x=7:
    f(7) = 83.46 (3 marks)
12. Formula (2 marks)
    Initial interval [0, 1] (since f(0)=1 >0, f(1)=−7 <0)
    1st approximation: 0.5 (2 marks)
    2nd approximation: 0.25 (2 marks)
13. Correct graph(2 marks)
    Corner Points: (0,0), (0,450), (150,300), (300,0) (2 marks)
    Optimal Solution: (0,450) → Z = 1800 (2 marks)

14. Let x = dolls, y = cars
    Maximize Z = 15x + 20y (1 mark)
    Subject to: $x + 3y \leq 90$ (Assembly) (1 mark)
    $2x + y \leq 80$ (Painting) (1 mark)
    $x,y \geq 0$
    Simplex Solution: Optimal at x=30, y=20 → Z = 850 (3 marks)
    (stage wise marks can be set by the examiner)
15. First allocation table (2 marks)
    Rest of the steps (2 marks)
    Final Allocation (2 marks)

Origin	**D_1**	**D_2**	**D_3**
O_1	40	0	0
O_2	0	0	30
O_3	0	20	10

16. First allocation table (2 marks)
    Rest of the steps (2 marks)
    Final Allocation table (2 marks)

Origin	**D_1**	**D_2**	**D_3**	**Supply**
O_1	**50**	0	**150**	200
O_2	**40**	**120**	0	160
O_3	**90**	0	0	90
Demand	180	120	150	

Part C

Answer any 2 questions. Each question carries 15 marks

17. (a) Euler Cycle – Correct definition (3 marks)
    Origin - famous Königsberg Bridge Problem solved by Euler in 1736. (3 marks)
    Conditions for Euler Cycle - A connected graph has an Euler cycle if: The graph is connected & Every vertex has even degree. (4 marks)
    (b) Constructing Euler graph and identifying Euler cycle (5 marks)
18. Difference table (7 marks)
    Formula (3 mark)
    Rest of the steps & $\sin 25° \approx 0.4226$ (5 marks)
19. Standard form (3 marks)
    Initial table (5 marks)
    Rest of the answer (4 marks)
    Optimal Solution: $x_1 = 0, x_2 = 0, x_3 = 10$, Maximum Z = 30 (3 marks)
20. Initial Basic Feasible Solution obtained by NWCM
    (S1,D1) – 5, (S1,D2) – 2, (S2,D2) – 6, (S2,D3) – 3, (S3,D3) – 4, (S3,D4) – 14
    Transportation cost =1015 (6 marks)
    Full marks may be given to solution using NWCM (15 marks) **OR**
    Testing for optimality using MODI & rest of the steps – 2 iterations (6 marks)
    Final allocation table(3rd iteration) & total transportation cost = 743(3 marks)

(S1,D1) – 5, (S1,D4) – 2, (S2,D2) – 2, (S2,D3) – 7, (S3,D2) – 6, (S3,D4) – 12